

DevOps Project 6

Configuration Management Using Ansible

Goal

Use **Ansible** to automatically configure a **remote Linux server** (instead of installing software manually).

Think of Ansible as: “Write once → run everywhere → no manual work”

Clone the Source code : <https://github.com/prade01/ansible-project>

Tools and Technology Used:

- Two Linux VM (CentOS/Ubuntu)
- Git
- Ansible Installed on One VM to manage other
- Dependencies of Python2/3
- Vi Editor

Installing Ansible on Control Node/Master Server

```
vagrant@db01:~$ sudo apt install ansible -y
```

Checking Ansible Version on Master Server

```
vagrant@web01:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/vagrant/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jan  8 2026, 06:52:19) [GCC 11.4.0]
vagrant@web01:~$
```

Testing Connectivity between Control Node and Target Server VM

```
vagrant@web01:~/ansible-project$ ansible all -i inventory -m ping
web01 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
db01 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host web02 should use /usr/libexec/platform-python, but is using /usr/bin/python for
backward compatibility with prior Ansible releases. A future Ansible release will default to using the discovered platform python
for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
web02 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host db02 should use /usr/libexec/platform-python, but is using /usr/bin/python for
backward compatibility with prior Ansible releases. A future Ansible release will default to using the discovered platform python
for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This
feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
db02 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
}
```

Run the ansible playbook to install & Start the Nginx Server on one Target VM

```
vagrant@web01:~/ansible-project$ ansible-playbook -i inventory setup.yaml

PLAY [Install and Start Nginx on Ubuntu and CentOS] *****

TASK [Gathering Facts] *****
ok: [db01]

TASK [Install Nginx on Ubuntu] *****
changed: [db01]

TASK [Install Nginx on CentOS] *****
skipping: [db01]

TASK [Start and Enable Nginx] *****
ok: [db01]

PLAY RECAP *****
db01                : ok=3    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

Check the Status of Installed Nginx Server on Target VM using Ansible

[illegible]

